

Body Controller Front

W,U X010 X011 X012 X013 X014 X015 X016 X017 X018 X019 X020 X021 X022

X021-1 Vref Output
Refrigerant Discharge SV
X021-2 Analog In
Refrigerant Discharge Pressure
X021-3 Analog In
Refrigerant Discharge Temp
X021-4 Ground Return
Refrigerant Discharge GND

X021-5 Ground Return
Refrigerant Suction GND
X021-6 Analog In
Refrigerant Suction Temp
X021-7 Analog In
Refrigerant Suction Pressure
X021-8 Vref Output
Refrigerant Suction SV

X021-33 Half-Bridge
Coolant Pump Battery Phase A
X021-32 Half-Bridge
Coolant Pump Battery Phase B
X021-31 Half-Bridge
Coolant Pump Battery Phase C
X021-34 Ground Return
Battery Coolant Pump GND Shield

X021-52 Half-Bridge
Coolant Pump Powertrain Phase A
X021-51 Half-Bridge
Coolant Pump Powertrain Phase B
X021-50 Half-Bridge
Coolant Pump Powertrain Phase C
X021-29 Ground Return
Powertrain Coolant Pump GND Shield

X021-19 Digital Out
Coolant Level Conductive Sensor
X021-40 Ground Return
Coolant Level Switch GND

X021-16 Analog In
Coolant Temp 1
X021-38 Ground Return
Coolant Temp 1 GND

X021-17 Analog In
Coolant Temp 2
X021-39 Ground Return
Coolant Temp 2 GND

X021-46 Half-Bridge
5-Way Coolant Valve B
X021-24 Half-Bridge
5-Way Coolant Valve A
X021-22 Half Sensor In
5-Way Coolant Valve Hall
X021-23 Ground Return
5-Way Coolant Valve Sensor GND

X021-25 Half-Bridge
HVAC Chiller EXV Phase A
X021-47 Half-Bridge
HVAC Chiller EXV Phase B
X021-26 Half-Bridge
HVAC Chiller EXV Phase C
X021-48 Half-Bridge
HVAC Chiller EXV Phase D

X021-15 High-Side Drive
HV Compressor Controller LV Power
X021-21 Pass-through
Vehicle CAN H
X021-43 Pass-through
Vehicle CAN L
X021-37 Ground Return
HVAC Compressor Controller LV GND

SH_TH010-A
TWIST_TH010-A

SHIELD895

SH_TH020-A
TWIST_TH020-A

SHIELD896

TWIST_CVH01_TH-A

TH001-A GN 0.13 MCV
B12.0 1068767.XCF (U)
TH002-A GY 0.13 MCV
B12.0 1068767.XCF (U)
TH003-A WH 0.13 MCV
B12.0 1068767.XCF (U)
TH004-A BK 0.13 MCV
B12.0 1068767.XCF (U)

TH005-A VT 0.13 MCV
711.0 1068767.XCF (U)
TH006-A GN 0.13 MCV
711.0 1068767.XCF (U)
TH007-A BU 0.13 MCV
711.0 1068767.XCF (U)
TH008-A BK 0.13 MCV
711.0 1068767.XCF (U)

TH011-A RD 1.00 FLRY
1154.0 1068767.XCF (U)
TH012-A YE 1.00 FLRY
1154.0 1068767.XCF (U)
TH013-A BU 1.00 FLRY
1154.0 1068767.XCF (U)

TH021-A GN 1.00 FLRY
588.0 1068767.XCF (U)
TH023-A WH 1.00 FLRY
588.0 1068767.XCF (U)
TH023-A VT 1.00 FLRY
588.0 1068767.XCF (U)

TH031-A BU 0.35 FLRY
457.0 1068767.XCF (U)
TH032-A GY 0.35 FLRY
457.0 1068767.XCF (U)

TH033-A VT 0.35 FLRY
919.0 1068767.XCF (U)
TH034-A WH 0.35 FLRY
919.0 1068767.XCF (U)

TH035-A YE 0.35 FLRY
460.0 1068767.XCF (U)
TH036-A RD 0.35 FLRY
460.0 1068767.XCF (U)

TH042-A GN 0.13 MCV
1229.0 1068767.XCF (U)
TH041-A RD 0.13 MCV
1229.0 1068767.XCF (U)
TH044-A BK 0.13 MCV
1229.0 1068767.XCF (U)
TH043-A GY 0.13 MCV
1229.0 1068767.XCF (U)

TH045-A YE 0.13 MCV
757.0 1068767.XCF (U)
TH046-A GY 0.13 MCV
757.0 1068767.XCF (U)
TH047-A BU 0.13 MCV
757.0 1068767.XCF (U)
TH048-A BK 0.13 MCV
757.0 1068767.XCF (U)

TH051-A BU 0.35 FLRY
688.0 1068767.XCF (U)
+CVH01_THA WH 0.35 FLRY
688.0 1068767.XCF (U)
+CVH02_THA YE 0.35 FLRY
688.0 1068767.XCF (U)
TH052-A GY 0.35 FLRY
688.0 1068767.XCF (U)

HV Compressor
U,W X530

X530-1 Power In
Power
X530-2 CAN
CAN HIGH
X530-3 CAN
CAN LOW
X530-4 Ground Return
GND
X531-1 Analog In
Analog In

Refrigerant Temp and Pressure Sensor - Discharge

X515-4 Power In
Vref
X515-2 Analog Out
Press
X515-3 Analog Out
Temp
X515-1 Ground Return
Ground

Refrigerant Temp and Pressure Sensor - Suction

X514-1 Ground Return
Ground
X514-3 Analog Out
Temp
X514-2 Analog Out
Press
X514-4 Power In
Vref

Battery Coolant Pump

X508-1 Power In
Phase 1
X508-2 Power In
Phase 2
X508-3 Power In
Phase 3
X508-4 Ground-Chassis
GND

Powertrain Coolant Pump

X510-1 Power In
Phase 1
X510-2 Power In
Phase 2
X510-3 Power In
Phase 3
X510-4 Ground Return
GND

Coolant Level Switch

X512-1 Analog Out
Capacitive Sensor
X512-3 Ground Return
GND

Coolant Temp Sensor - Battery

X517-1 Analog Out
Sensor In
X517-2 Ground Return
GND

Coolant Temp Sensor - Powertrain

X518-1 Analog Out
Sensor In
X518-2 Ground Return
GND

5-way Valve

X520-1 Half-Bridge
Motor +
X520-2 Half-Bridge
Motor -
X520-3 Hall Sensor Out
VBatt
X520-4 Ground Return
Ground

Chiller EXV

X521-1 Half-Bridge
Stepper Motor A IN
X521-3 Half-Bridge
Stepper Motor B YF
X521-2 Half-Bridge
Stepper Motor C XB
X521-4 Half-Bridge
Stepper Motor D XF

TESLA

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DRAWN		CHECKED		APPROVED	
MATERIAL		FINISHING		SCALE	
AS SPECIFIED		AS SPECIFIED		NONE	
PART NAME					
HVAC Thermal Subassembly					
PART NUMBER		REV	SHEET	SIZE	
		2	28 of 45	A0	